

Customer Shopping Analysis

- Data Analysis Project
- Python, SQL Server, Power BI

Business Problem

Analyze customer shopping behavior to identify purchasing patterns, customer segments, revenue drivers, and opportunities for improving customer retention and sales performance.

Dataset Information

- Records: 3,900
- Features: 18
- Customer demographics
- Purchase history
- Payment methods
- Shipping preferences
- Product categories
- Customer reviews

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually
6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	Venmo	Weekly

Data Cleaning & Feature Engineering

Data Cleaning

- Renamed columns
- Standardized column names
- Missing value treatment
- Duplicate removal

Missing review ratings were imputed using category median ratings to preserve category-specific patterns.

Feature Engineering

- Binning was used to group ages into specific age categories (Adult, Middle Age, Men, Senior), with the aim of simplifying the analysis and making comparisons between different age groups and discovering the group that spends the most or buys the most.
- Purchase frequency data was stored in text categories such as Weekly, Monthly, and Annually, so I performed Feature Engineering by converting these values to the equivalent number of days for each period (Weekly = 7 days, Monthly = 30 days, Annually = 365 days) so that the data could be mathematically and analytically processed and more accurate indicators of customer behavior could be extracted.

SQL Analysis

I conducted some queries to answer some essential questions that would benefit the business in understanding the data and obtaining more information about the buyers Including the following:

What is the total revenue generated by male vs. female customers?

	gender	total revenue
1	Male	157890
2	Female	75191

Which are the top 5 products with the highest average review rating?

	item_purchased	avrage rating
1	Gloves	3.861
2	Sandals	3.844
3	Boots	3.819
4	Hat	3.801
5	Skirt	3.785

Which customers used a discount but still spent more than the average purchase amount?

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60

✓ Query executed successfully.

Compare the average Purchase Amounts between Standard and Express Shipping.

	shipping_type	average
1	Standard	58
2	Express	60

Compare average spend and total revenue between subscribers and non-subscribers.

	total_customers	subscription_status	avg_spend	total_revenue
1	2847	No	59	170436
2	1053	Yes	59	62645

Which 5 products have the highest percentage of purchases with discounts applied?

	item_purchased	discount_percentage
1	Hat	50.000000
2	Sneakers	49.655100
3	Coat	49.068300
4	Sweater	48.170700
5	Pants	47.368400

Segmenting customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

	customer_segment	count of customers
1	Loyal	3116
2	Returning	701
3	new	83

customers who are repeat buyers more than 5 previous purchases

	subscription_status	number of buying
1	Yes	958
2	No	2518

What is the revenue contribution of each age group?

	age_groub	total revenue
1	Men	67353
2	Middle Age	65842
3	Senior	65256
4	Adult	34630

Power BI Dashboard

KPIs

- Total Revenue
- Total Customers
- Average Rating
- Number of Cities

Sclicer by

- Gender
- Age Group
- Season
- Category
- Shipping Type
- Discount Applied



Key Findings

- Clothing category generated the highest revenue.
- Subscribers represent 27% of customers and contribute approximately 27% of revenue. Therefore, there is no significant difference in purchasing power between subscribers and non-subscribers.
- Montana generated substantially more revenue than other cities.
- In the fall season, the highest revenue.
- Female customers exhibit slightly higher average spending per transaction compared to male customers (60.25 vs 59.54). However, male customers contribute more to total revenue due to their larger population share, indicating that revenue differences are driven primarily by customer distribution rather than spending behavior.